# Circular motion

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Relate inward net force to a change in velocity direction.

**SUCCESS CRITERIA - I can...**

1. Draw tangent velocity and inward acceleration.
2. Use $v^2/r$ or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Two balls launch horizontally from the same height at 3 and 15 m/s. Ignore air resistance. Compare fall times and horizontal ranges.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

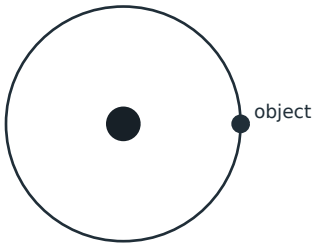
**Worked reasoning:** A 2 kg object moves at 3 m/s in a circle of radius 3 m. Inward acceleration is $v^2/r=3$ m/s²; net force is 6 N inward.

**Your turn:** Draw tangent velocity and inward net force at two different points.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

Add velocity and acceleration arrows to the marked point on the circle. Name one interaction that could provide the inward force.




The center is the attracting body; the small dot on the circle is the orbiting object.

**Marzano check - circle:** ① ② ③

## 04 Build the solution

A 2 kg object has supplied inward acceleration 3 m/s². Find inward net force. If acceleration doubles, what happens to force?

**Marzano check - circle:** ① ② ③

# Circular motion

**antocciphysics**



Continue down each column.

### 05 Use the idea

A string breaks while a ball moves in a circle. Neglect other forces over the next instant. Sketch its initial path after release.

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A 8 kg cart starts from rest under 80 N net force right for 7 s. Find acceleration and final velocity. State one assumption.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A car turns at steady speed. Can net force be zero? Explain using velocity.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Return to the worked example in problem 2. At the same radius and mass, compare forces for speeds 3 and 6 m/s. Why is the answer not just twice as much?

**Marzano check - circle:** 1 2 3